String Matching (1)

Peerapon S.

CPE 231 Algorithms

Topics

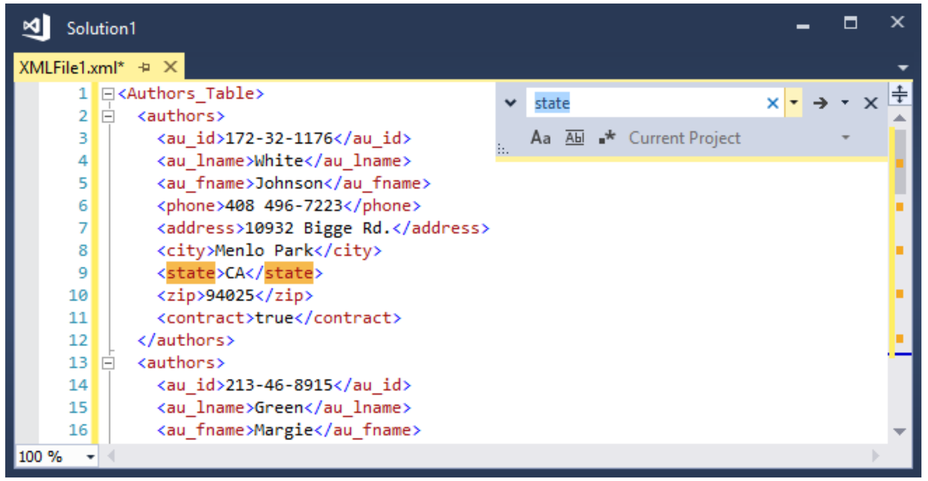
Brute-force string matching

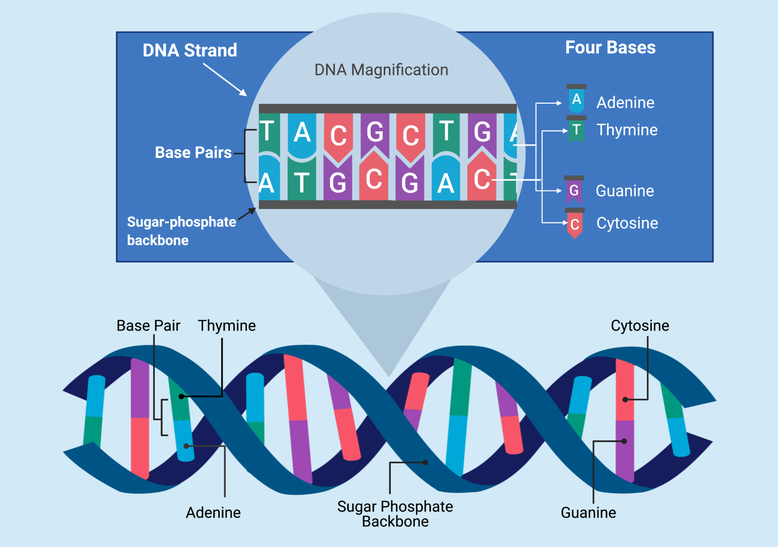
Horspool's algorithm

you can test, and prove or disprove. Social science is little more than observation putting on airs' (Kinsley, 2009).

Yet the invocation of an acronym, STEM, to group together science disciplines, and even more overtly the antonymic and ideological terminology of 'hard' and 'soft' disciplines, both speak to the decreasing usefulness of the idea of 'natural' or 'physical' sciences. In an increasingly human-made world, in what sense are the subject matters of engineering, medicine, dentistry, agricultural science or modern mathematics concerned only with 'natural' or even 'physical' systems? In what ways too are mathematical or quantitative social sciences such as econometrics or actuarial science any less 'hard' than biology or zoology? How is a randomized control trial carried out in social work or public management any less 'hard scientific' than one in medical pharmacology?

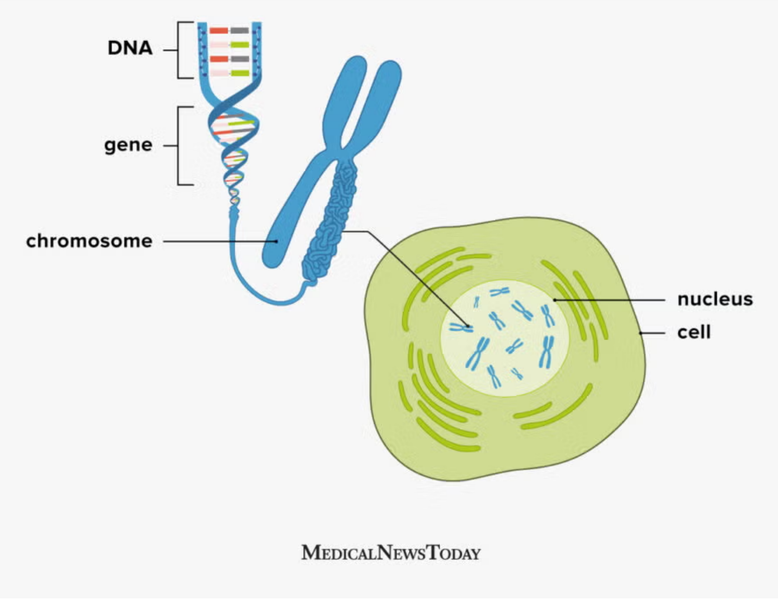
The mathematicization, quantification, formalization and theorization of the social sciences are still very partially advanced, but they are clearly the intellectually dominant trends in most disciplines – although the first three shifts are contested bitterly by a still predominant rear-guard of 'constructivists' opposed to any 'normal science' or 'positivist' model of the social sciences. But the impact of successive waves of scientific advances and fashions have made evident changes to the standards of what counts as 'evidence' in every social science discipline. In 1995 the biologist Edward O. Wilson could still lament in highly critical language the persistence of foundational disputes in social science: 'A great many [scholars] even enjoy the resulting overall atmosphere of chaos, mistaking it for creative ferment' (Wilson, 1995: 182). Yet constructivist critiques have shifted character in all the social sciences in the last decade, only infrequently now decrying the use of organized empirical evidence. Instead they emphasize the need for multiple sources of evidence, multiple methods of study, a focus on holistic phenomena, close attention to meanings as well as behaviours, and frequent triangulation of different kinds of evidential information.





A, C, G, T

```
AAGCTTTACG CCTGTTCAGA AGGCATTGAT GCGGGTTGCG CGCACGTACC
ATTTAAGGAT AAGGAGAGAG AGAGAGAGAG AGAGAGCAGC ATTTGAAAGT
CAATTCTGTA ATCATCTCCA AAACAAAAGG AGCGATGGTT GACCAATCAG
AACGGAAAGT GAAAAAGACC ACCCCTCATT ACTTTTCACT CCCCCACAAA
AAAAAAATCT AGTTCTTGAT CGCTATTCGA AGAGATTGAG CCAAGCTGAT
TGAACTAATT ATCATAGCTT TGAAGATTTT TTTAAAAAAT ATGGATCACC
TCAACATAAC CCCACTATAA AGATGCATTG CACATTCACT TGACTCATCG
ACAACCGGCA CGGCTGCTGT AGATTTCTGG CGGCTAGCAG GGCGTGTGTA
TACCTCTAGT ATCCTGACAC AATGTATAGC CTCGACGCTA CATCTGCTTC
CGGACATAGA GCTCGCTAGC GAGTGGCCGG TTTCCCACGT ACGTTCGAGT
CTGGATTACA TGCTGTCTAG CAGTTCCACA TACAGAATGC GTCGTATGAG
TCAGAGTTCC AATGCGTACG TAAACAGTGA GTACGCCTGC AGCCCCGACT
CTCTCGCCTC GTCTCTCTAT CCCTACGTCC AAATTGACAT AATGACACAA
ATACCACCAT TATATGGACT AGAATTCGTA GTAGCAATTT AGTAGACCCC
AAGTTGACCA TATATAAATA AGAGAAAATC AAGATCGTTT CAGATTCGAA
TCTGGTTAGC AGTACATACT ATTGGGTTTC CTTTGATGTG TATGACATCT
CAGGGACCTT GCTTTTCACA TGCAGTTTCC CAGCATCCTC AGTTATCTTG
GCTGATTTGG ACATGCTTGC AAGTTAATTG AAATTTATGA GCTTGATTGC
CATTTGCTTG ATAATTGGTA GCTGGGACCA AATCTGAAGT GATTGCATTT
CAGGCAACAT AGGAAGGTCC CTTTCTATGC TGCTCTACCC CCCAGGCAAT
ACTAGAAATA TCTGTGTACA AGTCACACAT GTTTACATAT TTATTTCTTC
AATTTCTATC TCACCTTTCC TCCCCAAACT CCAGGCAGCC TATATTTGGG
AACTGGCAAT CTATGCATTT GCCAGCAAAT AAATGCAATG CATGGTCCAT
TGAATAAAAT CCCCATTTTT CTCAAAAATT TCCAGGACTC ATATCTGACT
GTAAATTGTG CATCCAGGAA GCAATTGCTG ATTTCAAAAT ATGTGGTATA
CAGCAATCAA CTGTAAAGAG AATTACAGTT TAGGTTACAT CATAAATGCA
CATTTGTTGC TCAGCTGAAT CAAGAGGTAG GTGATCTTCA TGTGTGAATG
GGTTACTGGA ACAAGTAGTT ATACAGTAAA CCCACTGAGA AACAGGCAAG
TAAAATAAAT GTCAAGAACC TGATCTAAGA CATATTTGAA AAACAGAAGT
GATCAATGAA AAGAACTACA TGTCTCCAAA TGTACATTAT TTTTTCAATC
GCATTATTTC AGATTTCACA GTTTTCCCTT CAACTAATAA AAATGGAATA
CTTGTGGAGA CAGTCCACAG TCTATTATAA TAGCACAATG GTGAAGAAAA
AAAATGATAG TCTCATTTTG AGCAAATTAC TTATTTAATC TATTACTGTA
TTATACTGTA TTTAATTTGG AGGTCATCTA ATCCAACCTC AGGCTCTGAA
GGAAAACATA CCAATTAATG TAGCAGCCTA TACTACTGGC TAATGGTTCA
GACAGGAAGT TCCTTTTGAC ATCTAGCTTG GAATTACTTC CTGAAATTTT
TAACCTGTTC CTGCCCTTTA GGTTAGCTGA GAATAAGACC CCACTGTTTG
TCTATGTGAC AAGCTT
```




```
  1 GGACCTGGAATATGGCGAGAAAACTGAAAATCACGGAAAATGAGAAATAC
 51 ACACTTTAGGACGTGAAATATGGCGAGGAAAACTGAAAAAGGTGGAAAAT
101 TTAGAAATGTCCACTGTAGGACGTGGAATATGGCAAGAAAACTGAAAATC
151 ATGGAAAATGAGAAACATCCACTTGACGACTTGAAAAATGACGAAATCAC
201 TAAAAAACGTGAAAAATGAGAAATGCACACTGAA 234
```

Ref: https://www.fjc.gov/content/361230/DNA-basics-nucleotides-genes-genome

Substring Search/Matching Problem

character/alphabet

Consider a string/pattern $P = p_0 p_1 \cdots p_{m-1}$ ($m = |P|$ characters) and text $T = t_0 t_1 \cdots t_{n-1}$ ($n = |T|$ characters, $n \geq m$)

P = A A C

T = G T A A C A G T A A A C G

$\Rightarrow \Sigma = \{A, C, G, T\}$

$\Sigma = \{A, B, C, D, \ldots\}$

$\Sigma, 1, 2, 3 \ldots \}$

Elements in P and T are characters drawn from a finite alphabet set Σ.

Goal: Determine if P appears in T. If so, where ?

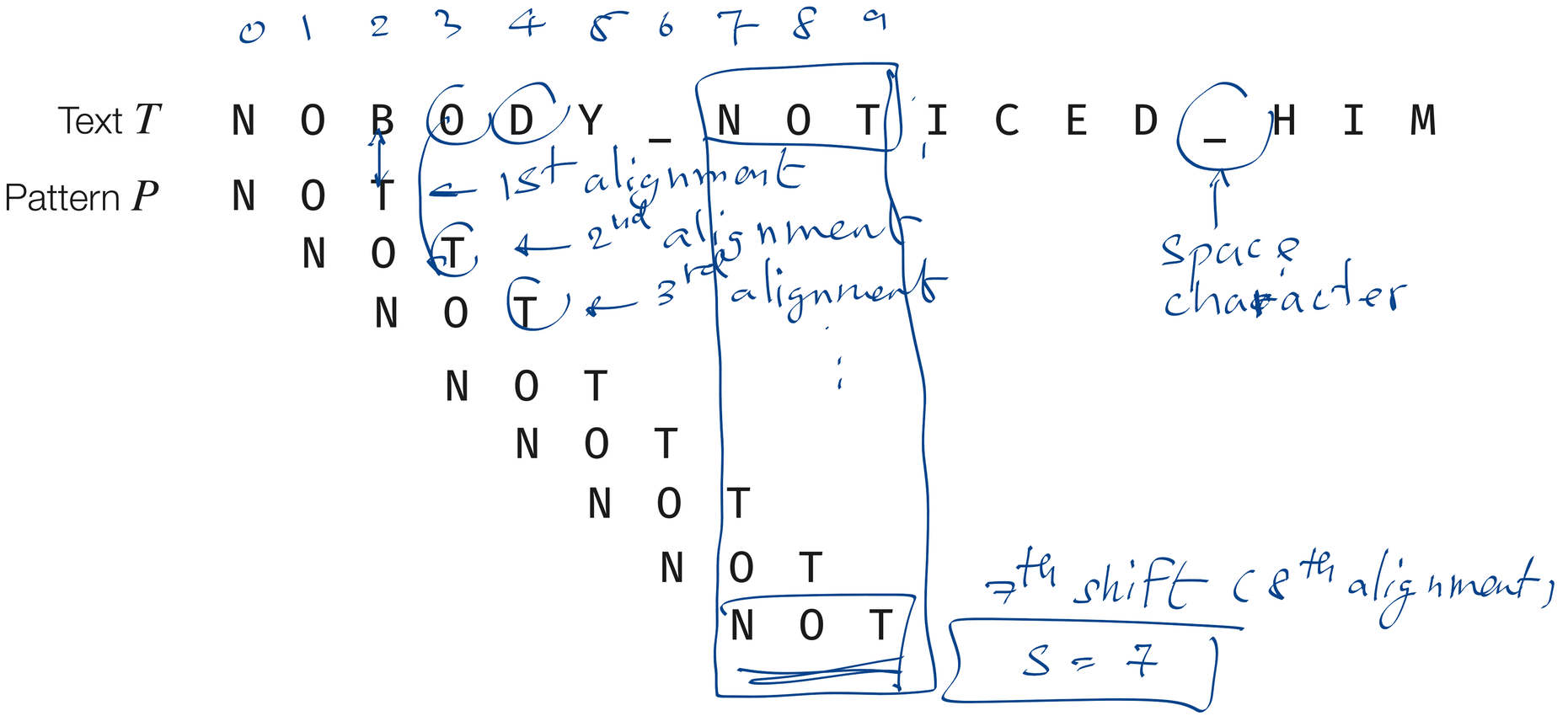
Solution: For $0 \leq s \leq n - m$, P occurs with shift s in T if $t_s = p_0, t_{s+1} = p_1, \ \ldots \ , t_{s+m-1} = p_{m-1}$

$s = 2$ 1st match

$s = 9$ 2nd match

Brute-Force String Matching

Align pattern to text, compare (left → right or right → left), and shift the pattern by one position if a mismatch is encountered.



$n = 18,\ m = 3$

Self Test: Brute-Force String Matching

https://forms.office.com/r/AqrC7nawe1?origin=lprLink

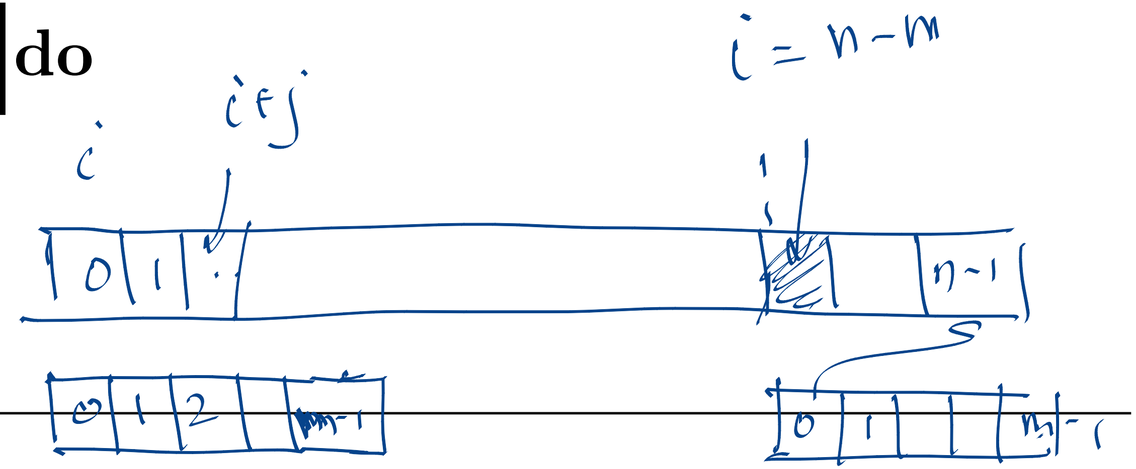


Algorithm BruteForceStringMatching

Input: Pattern $P = p_0 p_1 p_2 \cdots p_{m-1}$ and Text $T = t_0 t_1 t_2 \cdots t_{n-1}$, $n \geq m$
Output: Index of the 1st character in the text that starts a matching
substring, and -1 for the unsuccessful search

```
for i = 0 to n-m do
    j ← 0
    while j < m and p_j == t_{i+j} do
        j ← j + 1
    if j == m return i
return −1
```

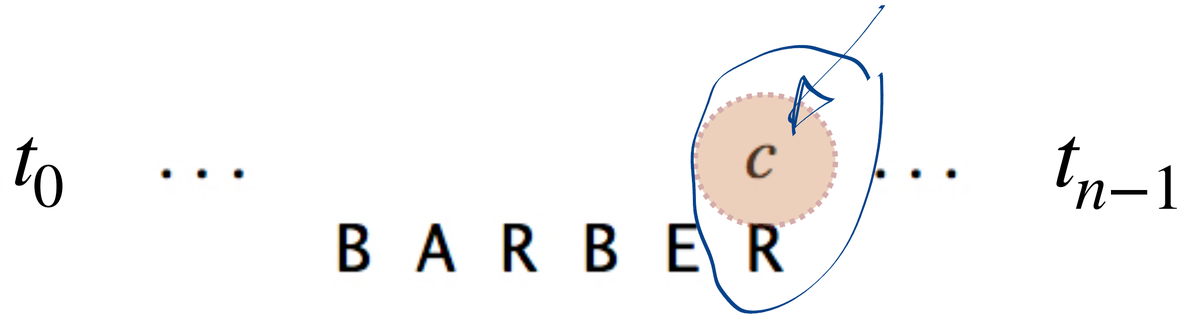
a match of p_j and t_{i+j}



$\Theta(n\,m)$ worst-case (pattern not in the text)

Horspool's Algorithm

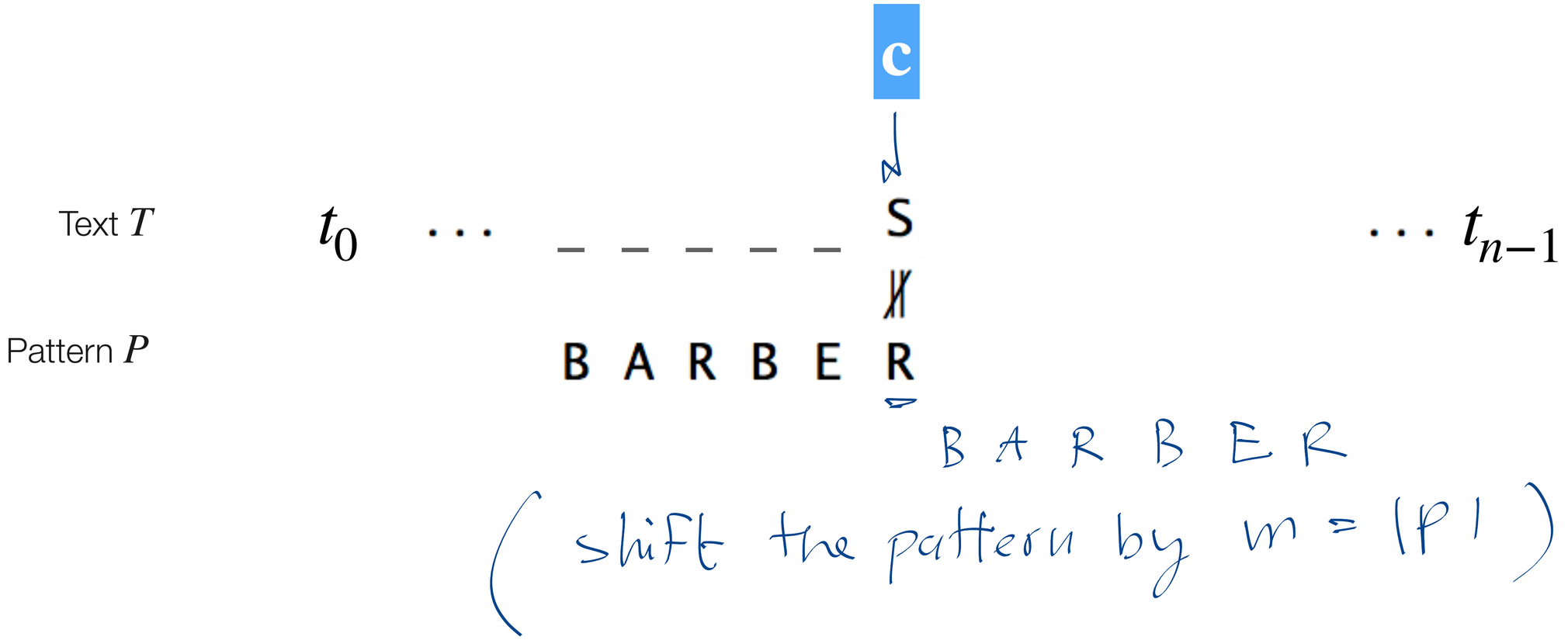
Ex: Search for pattern 'BARBER' ($m = 6$) in a text $T = t_0 t_1 t_2 \cdots t_{n-1}$.



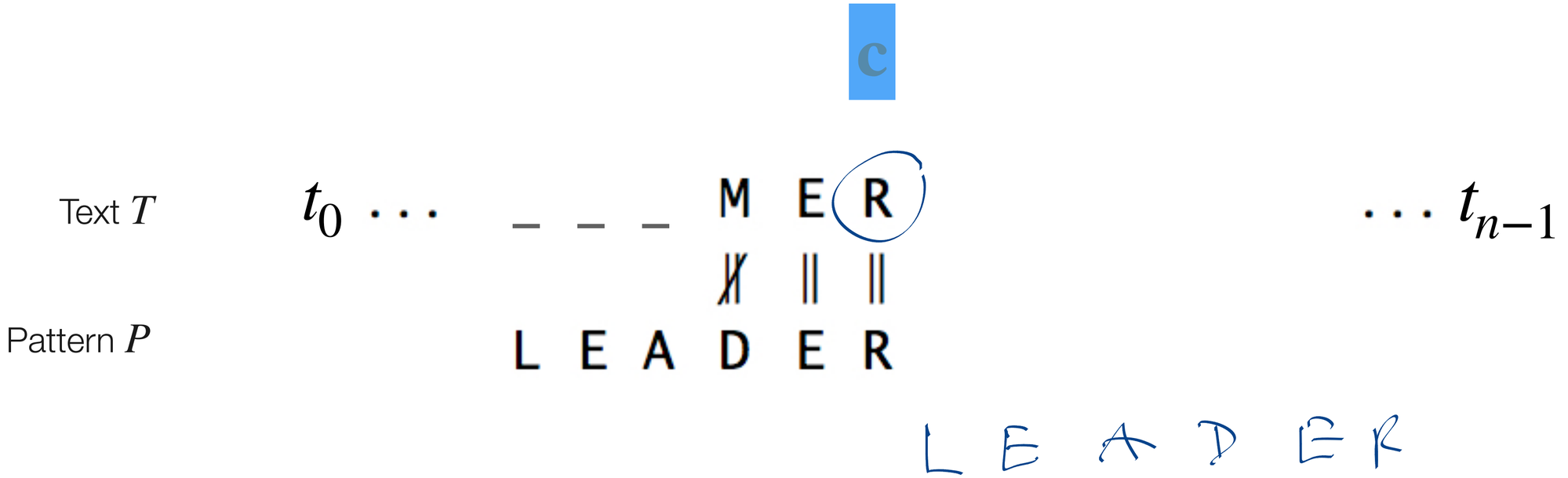
Compare the characters from ***right to left*** in the alignment ***until encountering a mismatch***.

Make a largest shift possible based on the ***last character c of the text aligning with the last character of the pattern.***

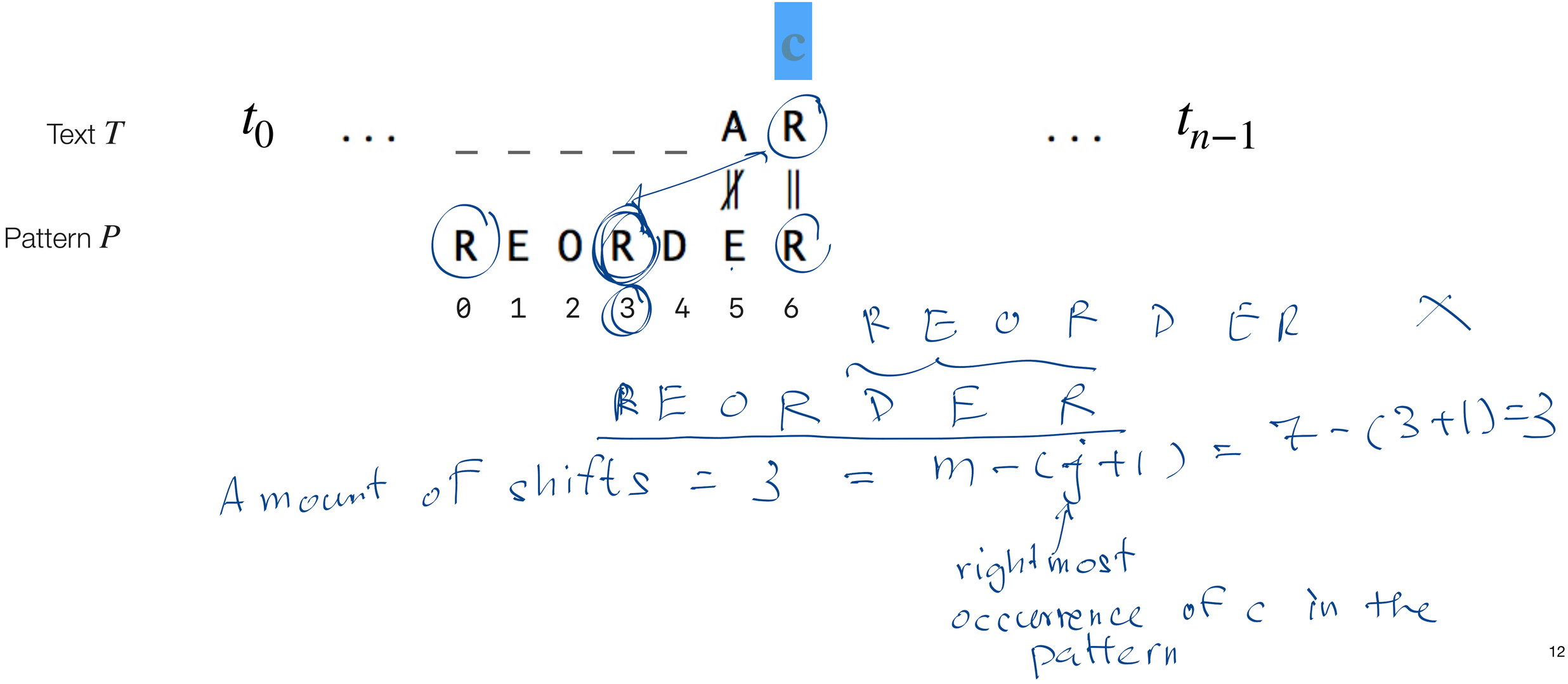
- **Case 1**: Character $c \in T$ is not in P.
  - 'S' mismatches 'R'
  - 'S' not in 'BARBER'



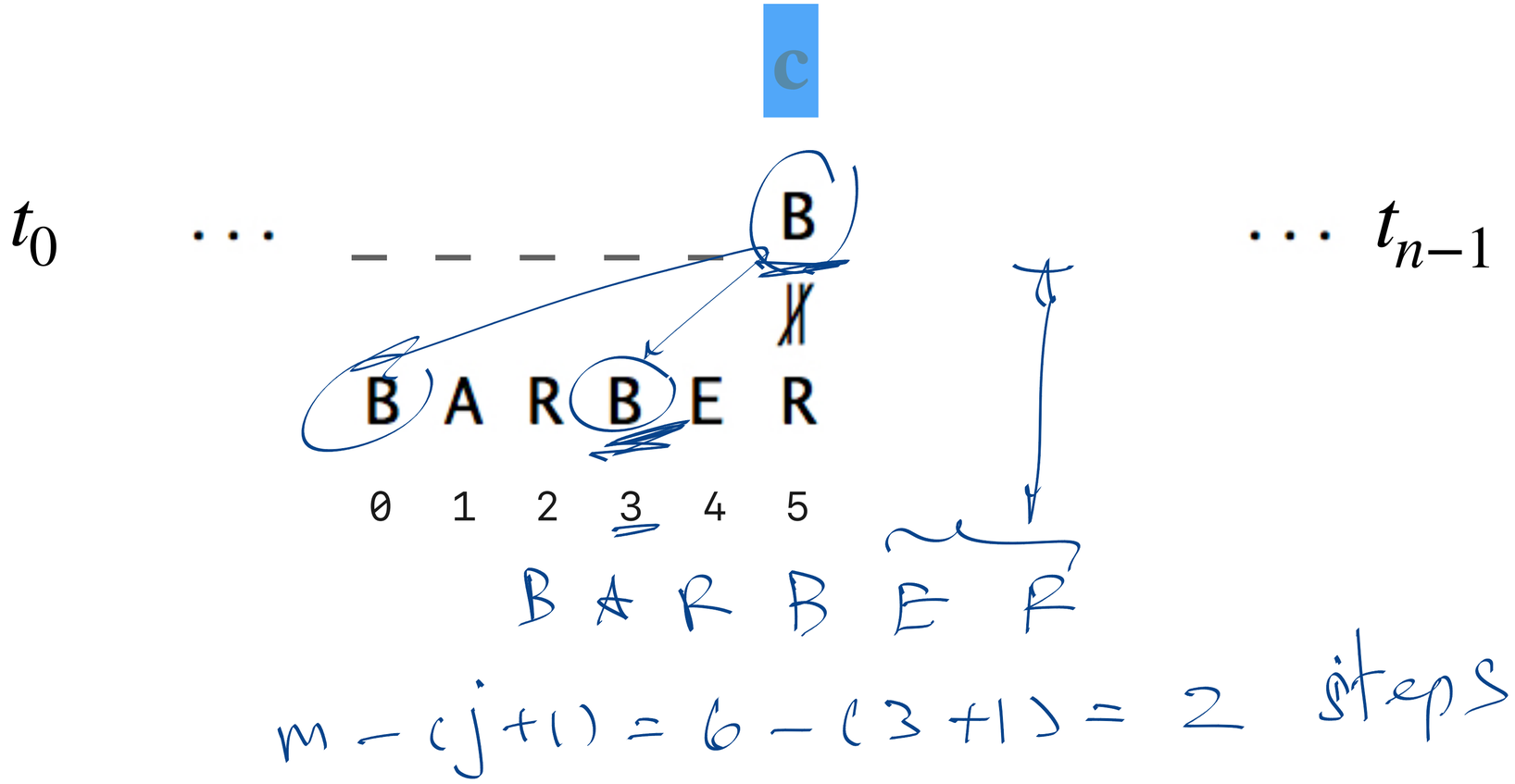
- **Case 2**: Character $c \in T$ appears only once in the last position of P.



- **Case 3:** Character $c \in T$ appears in the last position of P and also somewhere else in P.
  - c='R' matches 'R' but the alignment mismatches.
  - c='R' appears more than once in 'REORDER'



- **Case 4:** Character c appears in P but not in the last position.
  - 'B' mismatches 'R'
  - But 'B' appears in 'BARBER'



Inefficient to check the occurrence of character $c \in T$ in the pattern P every time the alignment mismatches.

Given a pattern, the shift size for each character c can be **precomputed** and **stored** in a table.

Assuming $\Sigma = \{A, B, C, \ldots, Z, _\}$

Shift table for pattern P = 'BARBER'

	0	1	2	3	4	5		18		26	27
character c	A	B	C	D	E	F	. . .	R	. . .	Z	_
shift $t(c)$	4	2	6	6	1	6	6	3	6	6	6

How to convert characters to indices of $t(.)$ in the shift table ?

Converting Character Indices to Numeric Indices

For Σ being a set of all printable characters and english alphabets,

Code	Char	Code	Char	Code	Char	Code	Char	Code	Char	Code	Char
32	[space]	48	0	64	@	80	P	96	`	112	p
33	!	49	1	65	A	81	Q	97	a	113	q
34	"	50	2	66	B	82	R	98	b	114	r
35	#	51	3	67	C	83	S	99	c	115	s
36	$	52	4	68	D	84	T	100	d	116	t
37	%	53	5	69	E	85	U	101	e	117	u
38	&	54	6	70	F	86	V	102	f	118	v
39	'	55	7	71	G	87	W	103	g	119	w
40	(	56	8	72	H	88	X	104	h	120	x
41	)	57	9	73	I	89	Y	105	i	121	y
42	*	58	:	74	J	90	Z	106	j	122	z
43	+	59	;	75	K	91	[	107	k	123	{
44	,	60	<	76	L	92	\	108	l	124	|
45	-	61	=	77	M	93	]	109	m	125	}
46	.	62	>	78	N	94	^	110	n	126	~
47	/	63	?	79	O	95	_	111	o	127	[backspace]

Printable english text characters range from 32 to 126 (size = 95)

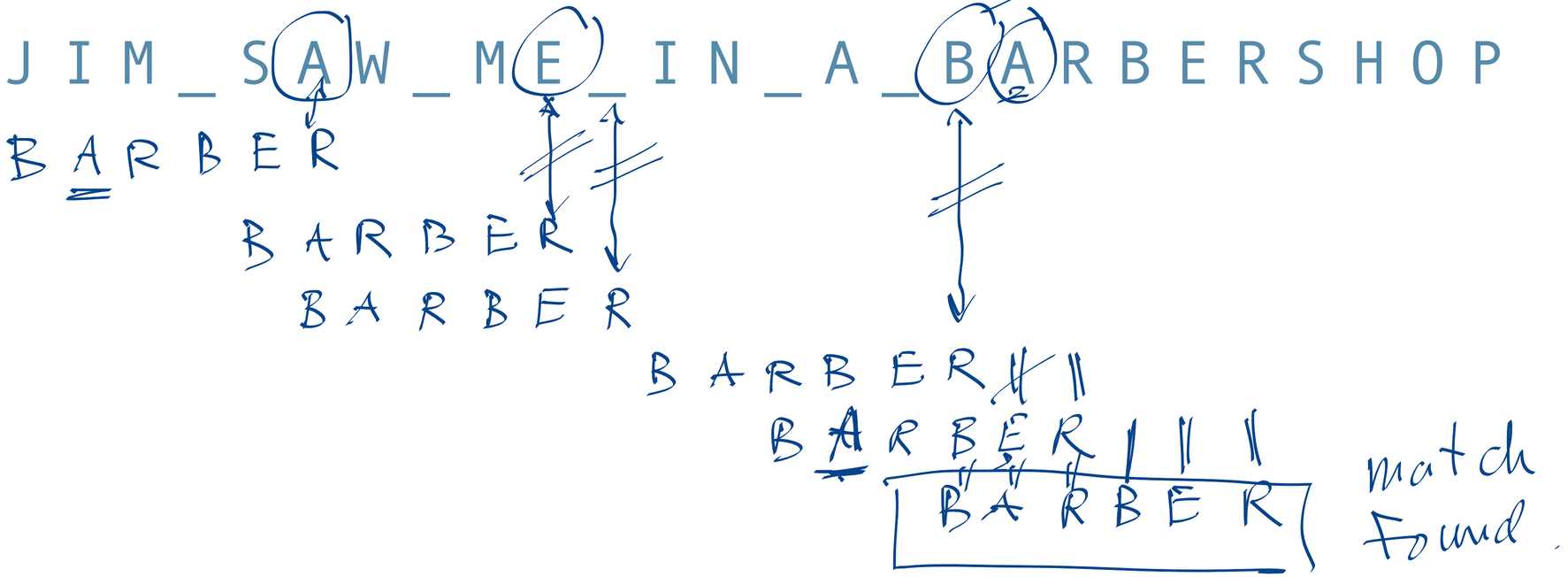
```
Space:  ' '-32 = 32-32 = 0
!:      '!'-32 = 33-32 = 1
...
A:      'A'-32 = 65-32 = 33
..
~:      '~'-32 = 126-32 = 94
```

Example

```
P = 'BARBER'
```

character c	A	B	C	D	E	F	. . .	R	. . .	Z	_
shift $t(c)$	4	2	6	6	1	6	6	3	6	6	6

Text: J I M _ S A W _ M E _ I N _ A _ B A R B E R S H O P



Constructing Shift Table

Let $\Sigma = \{A, B, C, \ldots, Z, _\}$ and pattern $P =$ BARBER, $P[i] = \text{ord}(p_i)$

$m = |P| = 6$

0 1 2 ... 26

Initialization

$P[0] = 1$ ('B')
$P[1] = 0$ ('A')
$P[2] = 17$ ('R')
$P[3] = 1$ ('B')
$P[4] = 4$ ('E')
$P[5] = 17$ ('R')

```
Table[0] = 6
Table[1] = 6
Table[2] = 6
...
Table[26] = 6
```

Update the shift table

```
Table[P[0]] = 6-1-0 = 5
Table[P[1]] = 6-1-1 = 4
Table[P[2]] = 6-1-2 = 3
Table[P[3]] = 6-1-3 = 2
Table[P[4]] = 6-1-4 = 1
```

'B' A R B E

$m - (j+1) = m - 1 - j$

update from 5 to 2

updated from 6 to 3

Shift size for character 'c' computed as

$$t(c) = \begin{cases} \text{the pattern's length } m, \\ \text{if } c \text{ is not among the first } m-1 \text{ characters of the pattern;} \\ \\ \text{the distance from the rightmost } c \text{ among the first } m-1 \text{ characters} \\ \text{of the pattern to its last character, otherwise.} \end{cases} \quad \textbf{(7.1)}$$

ALGORITHM *ShiftTable*$(P[0..m-1])$

ithms

```
//Input: Pattern P[0..m − 1] and an alphabet of possible characters
//Output: Table[0..size − 1] indexed by the alphabet's characters and
//        filled with shift sizes computed by formula (7.1)
for i ← 0 to size − 1 do Table[i] ← m
for j ← 0 to m − 2 do Table[P[j]] ← m − 1 − j
return Table
```

Initialization